

CHILLICOTHE 22 ENE, MO, USA

WMO: 725550

Lat: 39.8668N

Lon: 93.1470W

Elev: 254

StdP: 98.31

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 13301

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.1	-15.0	-22.5	0.5	-17.8	-20.1	0.6	-14.7	8.5	-1.9	7.9	-3.4	2.1	N/A	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	33.8	24.5	32.2	24.5	30.8	23.6	26.3	31.5	25.4	30.0	24.6	29.2	3.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.6	30.1	24.0	19.5	28.8	23.1	18.5	27.6	83.6	31.7	79.5	29.9	75.5	29.2	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.4	5.7	-22.4	35.6	3.0	2.0	-24.5	37.0	-26.3	38.1	-28.0	39.2	-30.2	40.6		
(5)			-23.3	27.6	3.2	1.1	-25.6	28.3	-27.5	29.0	-29.2	29.6	-31.5	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	-2.8	-1.5	6.3	11.8	17.4	22.7	24.4	23.5	19.7	12.6	5.8	-0.5		
	DBStd	10.96	6.84	6.89	6.46	5.36	4.63	3.07	3.05	2.95	4.25	5.49	5.76	6.04		
	HDD10.0	1429	397	326	152	41	4	0	0	0	35	148	327			
	HDD18.3	3053	654	555	376	201	74	4	1	34	193	377	583			
	CDD10.0	2047	1	4	36	97	233	381	446	418	292	114	22	2		
	CDD18.3	628	0	0	1	6	46	134	189	161	76	14	1	0		
	CDH23.3	5640	0	0	7	60	349	1206	1859	1418	626	114	1	0		
	CDH26.7	1985	0	0	0	7	74	406	734	536	203	24	0	0		
(14) Wind	WSAvg	2.5	3.0	3.1	3.3	3.2	2.4	2.2	1.9	1.8	2.0	2.4	2.6	2.7		
(15) Precipitation	PrecAvg	997	34	44	67	101	126	139	119	101	101	73	56	40		
	PrecMax	1456	74	104	133	173	300	286	391	269	277	199	175	108		
	PrecMin	643	5	3	12	30	26	21	21	1	17	10	0	0		
	PrecStd	208	19	26	33	39	73	62	83	60	66	52	44	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	20.3	24.4	27.7	31.0	34.0	36.6	35.5	33.1	29.3	22.3	17.0		
		MCWB	10.3	11.7	17.4	19.3	22.4	25.0	23.3	22.8	23.4	21.2	15.2	13.6		
	2%	DB	12.2	16.1	22.0	25.2	28.7	32.0	34.0	33.5	30.9	26.5	19.8	13.2		
		MCWB	8.0	10.9	15.1	17.1	21.5	24.2	24.9	23.8	23.2	18.8	13.8	9.8		
	5%	DB	9.8	12.0	19.4	23.0	26.9	30.7	32.3	31.6	29.3	24.0	17.2	10.5		
		MCWB	6.2	8.4	13.9	15.8	20.3	23.5	25.2	23.8	22.3	17.5	12.4	7.7		
	10%	DB	6.7	8.5	16.5	20.9	25.3	29.1	30.6	29.7	27.0	21.5	14.7	8.1		
		MCWB	4.2	5.2	11.4	14.8	19.1	22.8	24.1	23.2	21.0	16.4	10.3	5.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	14.4	18.8	20.3	24.0	26.5	27.4	27.4	24.9	22.4	16.8	15.3		
		MCDB	14.4	18.6	23.6	26.3	29.6	31.7	32.6	32.1	31.1	26.9	20.1	17.4		
	2%	WB	8.3	11.1	16.3	18.3	22.5	25.5	26.6	26.0	24.0	20.5	14.7	11.0		
		MCDB	10.9	14.9	20.5	23.4	27.4	30.4	31.9	30.4	29.8	24.1	18.0	12.7		
	5%	WB	6.1	8.5	14.5	17.0	21.4	24.6	25.7	24.8	23.0	18.9	12.6	8.3		
		MCDB	8.9	12.5	18.3	21.7	26.0	29.4	30.5	28.9	28.0	22.0	16.5	10.2		
	10%	WB	3.6	5.3	11.7	15.6	20.4	23.7	24.8	23.8	22.1	16.9	10.6	5.8		
		MCDB	6.2	8.4	15.3	20.2	24.6	28.6	29.2	28.0	25.9	20.7	13.8	8.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.0	10.6	11.1	10.3	10.2	10.4	10.5	11.4	11.1	10.6	8.7		
		MCDBR	13.4	15.2	13.8	13.8	12.1	11.7	11.7	12.7	12.3	13.2	13.6	11.7		
		MCWBR	9.7	10.2	8.2	8.2	6.3	5.7	5.1	5.3	5.2	6.8	9.2	8.7		
		MCDBR	12.5	14.3	11.7	11.8	10.8	10.6	9.9	10.1	10.7	10.4	12.2	10.6		
	5% WB	MCWBR	9.4	10.4	7.3	7.8	6.0	5.9	5.3	5.5	5.2	6.9	9.1	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.315	0.335	0.392	0.426	0.428	0.441	0.440	0.388	0.345	0.302	0.296		
	taud		2.443	2.379	2.404	2.258	2.232	2.259	2.223	2.245	2.342	2.456	2.517	2.464		
	Ebn at Noon		875	902	921	882	856	851	836	827	852	858	863	845		
	Edh at Noon		80	98	107	132	139	135	139	133	113	90	74	73		
(44) All-Sky Solar Radiation	RadAvg		2.00	2.87	3.76	4.86	5.66	6.32	6.33	5.53	4.76	3.27	2.24	1.75		
	RadStd		0.22	0.26	0.31	0.34	0.47	0.56	0.41	0.28	0.29	0.36	0.21	0.19		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page